

CHAPTER 1

INTRODUCTION

1.1 Introduction

Water is one of the most essential natural resources for sustaining life and maintaining ecological balance. However, rapid industrialization, urbanization, agricultural activities, and improper waste disposal have significantly contributed to the contamination of water resources worldwide. Water pollution has become a major environmental challenge, affecting both human health and aquatic ecosystems. Among the various indicators of water quality, turbidity is one of the most important parameters used to determine the presence of suspended particles such as soil, organic matter, microorganisms, and industrial pollutants. High turbidity not only reduces the aesthetic quality of water but also limits light penetration, affects aquatic life, and decreases the efficiency of conventional water treatment processes.

Traditional water purification methods commonly employ chemical coagulants such as alum and ferric salts for the removal of suspended particles. Although these chemicals are effective in reducing turbidity, their continuous use may produce harmful sludge, increase treatment costs, and introduce secondary environmental pollution. In recent years, researchers have focused on developing sustainable and environmentally friendly alternatives using naturally available biomaterials that possess excellent coagulation and adsorption properties.

Natural biomaterials such as **Moringa oleifera** seeds, banana peels, orange peels, and activated charcoal have attracted considerable attention due to their biodegradability, low cost, and high purification efficiency. Moringa seed proteins function as natural bio-coagulants by neutralizing negatively charged suspended particles, resulting in floc formation and sedimentation. Banana peel and orange peel powders contain cellulose, lignin, pectin, and various functional groups that enhance adsorption of contaminants through biosorption mechanisms. Activated charcoal possesses a highly porous structure and a large surface area, enabling effective adsorption of dissolved impurities and suspended particles.

To improve the handling and efficiency of these biomaterials, they can be encapsulated within **sodium alginate hydrogel beads**. Sodium alginate is a naturally occurring biodegradable polymer extracted from brown seaweed. When crosslinked with calcium chloride, it forms a three-dimensional hydrogel network through ionic gelation, commonly referred to as the "egg-box model." This hydrogel matrix provides mechanical stability, protects the encapsulated biomaterials, and facilitates controlled interaction between the active components and contaminated water. The integration of coagulation and adsorption mechanisms within a single biodegradable hydrogel bead enhances purification efficiency while maintaining environmental sustainability.

In addition to material development, modern water treatment systems increasingly utilize smart monitoring technologies for rapid assessment of water quality. The integration of an **ESP32 microcontroller** with a **turbidity sensor** enables real-time monitoring of turbidity levels, allowing comparative evaluation of water quality before and after treatment. Such smart monitoring systems improve data collection, facilitate automated analysis, and support the development of intelligent water management solutions.

The present project, titled "**Comparative Study of Multifunctional Biodegradable Bio-Coagulant Hydrogel Beads for Smart Water Treatment Applications**," focuses on developing environmentally friendly hydrogel beads using natural biomaterials for sustainable water purification. Different combinations of Moringa seed powder, banana peel powder, orange peel powder, and activated charcoal were encapsulated within sodium alginate hydrogel beads and comparatively evaluated for their turbidity reduction performance. A prototype-level IoT monitoring system based on an ESP32 microcontroller and turbidity sensor was incorporated to assess purification efficiency through comparative turbidity analysis.

This interdisciplinary project combines concepts from environmental biotechnology, polymer science, biomaterials engineering, and embedded systems to develop a low-cost, biodegradable, and sustainable water treatment solution. The study demonstrates the potential of agricultural waste-based biomaterials as eco-friendly alternatives to conventional chemical coagulants while promoting resource utilization, waste valorization, and environmental conservation.

Furthermore, the project supports the principles of green technology by reducing dependence on synthetic chemicals and encouraging the use of renewable natural resources. The proposed system has potential applications in rural water pre-treatment, educational demonstrations, environmental monitoring, and sustainable wastewater management. The integration of biodegradable purification materials with smart sensing technologies represents a promising approach toward achieving cleaner water and supporting the global objectives of sustainable development.

1.2 Scope

This project explores the use of biodegradable hydrogel beads prepared from natural materials for water quality improvement. The developed bio-beads can reduce turbidity and assist in the removal of contaminants through multiple treatment mechanisms, making them a sustainable and eco-friendly alternative to conventional methods. The system also integrates smart turbidity monitoring using an ESP32 and turbidity sensor for real-time assessment of water quality.

The project can be further enhanced by improving the bead formulation, incorporating additional natural adsorbents, and integrating advanced sensors for comprehensive water quality monitoring. Potential applications include wastewater pretreatment, irrigation water improvement, rural water management, and environmental remediation. The combination of biotechnology, sustainable materials, and IoT technology provides significant potential for future development in low-cost and environmentally friendly water treatment solutions. Augmented Reality turns the environment around you into a digital interface by placing virtual objects in the real world, in real-time. Augmented Reality can be seen through a wide variety of experiences. We distinguish three main categories of Augmented Reality tools.

1.3 Objectives

Current water treatment processes are largely dependent on synthetic chemical coagulants such as aluminum sulfate (alum), which, while effective, pose significant environmental and health

risks. The accumulation of metal-based sludge and the potential for secondary water contamination necessitate a shift toward sustainable, biodegradable alternatives.

While natural coagulants like *Moringa oleifera* show significant potential in treating water, their effectiveness is often evaluated through inconsistent manual observation. There is a need for a more standardized approach to quantify the performance of bio-based materials during the water treatment process. This project addresses these challenges by developing biodegradable hydrogel beads formulated with defatted *Moringa oleifera* powder. To ensure precise empirical validation, an ESP32-based turbidity monitoring system is utilized as a diagnostic tool, providing accurate, real-time tracking of NTU (Nephelometric Turbidity Unit) values to demonstrate the efficacy of the bio-coagulants in clearing contaminated water.

1.4 Organization of the IPBL project work

The remainder of this report is organized into the following chapters to guide the reader systematically through the development, engineering, and empirical evaluation of the multifunctional bio-coagulant hydrogel beads.

Research Objectives: This chapter establishes the core focus of the study, clearly outlining the specific goals for developing sustainable, biodegradable hydrogel beads. It emphasizes the integration of real-time smart monitoring systems, which serves as a technological bridge between traditional bio-coagulation methods and modern water quality management.

Literature Review: This section provides a comprehensive review of relevant scholarly work, focusing on the efficacy of natural coagulants such as *Moringa oleifera* and the rapid advancement of IoT-based water quality sensors. By synthesizing this information, the report highlights the current research gaps that this project aims to address through its unique combination of materials and smart technology.

Materials and Methodology: This chapter details the technical foundation of the study, describing the chemical composition of the hydrogel beads and the intricacies of the cross-linking process. It further explains the systematic development of the various experimental

formulations (B1–B4), ensuring that the fabrication process is transparent, reproducible, and scientifically grounded.

Smart System Design: This section describes the engineering and integration of the smart sensor system. It specifically explores the use of the ESP32 microcontroller and turbidity sensors, detailing how these components are configured to enable real-time performance tracking and data acquisition during the water treatment process.

Experimental Setup: This chapter outlines the rigorous experimental setup employed to test the beads. It details the preparation of simulated contaminated water, the standardization of pollutant levels, and the controlled environmental conditions under which the coagulation and adsorption tests were performed to ensure data integrity.

Results and Discussion: This chapter presents the empirical findings and provides a deep-dive comparative analysis. It evaluates the effectiveness of each bead formulation (B1–B4) in reducing turbidity, while simultaneously assessing the reliability and responsiveness of the integrated sensor data in reflecting water quality changes.

Conclusion and Future Scope: The final chapter summarizes the primary findings of the project, discussing the broader practical implications of the smart water treatment system for real-world applications. It concludes by reflecting on the project’s success and proposing tangible directions for future research and industrial-scale implementation.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Water pollution is a significant environmental challenge caused by rapid industrialization, urbanization, and agricultural activities. The presence of suspended solids, organic pollutants, heavy metals, and pathogenic microorganisms adversely affects water quality and poses serious health risks. Conventional treatment methods are often dependent on synthetic chemical coagulants, which may generate large amounts of sludge and increase operational costs. Therefore, recent research has focused on developing sustainable, biodegradable, and environmentally friendly alternatives using natural biomaterials (WHO, 2022; Benettayeb et al., 2022).

2.2 Water Pollution and Turbidity

Turbidity is an important physical parameter that indicates the concentration of suspended particles in water. High turbidity reduces light penetration, decreases the effectiveness of disinfection, and negatively impacts aquatic ecosystems. According to the World Health Organization (WHO), the turbidity of drinking water should preferably be below 5 NTU. Conventional treatment techniques such as coagulation, flocculation, sedimentation, and filtration effectively reduce turbidity but often require chemical coagulants like alum and ferric chloride, which may have environmental drawbacks (WHO, 2022).

Reference:

- World Health Organization (2022). *Guidelines for Drinking-water Quality*.

2.3 *Moringa oleifera* as a Natural Bio-Coagulant

Moringa oleifera seeds have been widely investigated as natural bio-coagulants due to the presence of water-soluble cationic proteins. These proteins neutralize negatively charged suspended particles through charge neutralization, resulting in floc formation and sedimentation. Studies have demonstrated that *Moringa* seed extracts effectively reduce

turbidity, colour, and microbial contamination while producing biodegradable sludge. Researchers have also reported that defatted Moringa seed powder exhibits improved coagulation efficiency because oil extraction increases the relative concentration of active coagulant proteins (Benettayeb et al., 2022; Baptista et al., 2017).

Research Papers

1. Benettayeb, A., et al. (2022). **Removal of Contaminants from Water Using Moringa oleifera as Biosorbent: An Overview of the Last Decade.** *Journal of Water Process Engineering.*
2. Baptista, A. T. A., et al. (2017). **Protein extraction from Moringa oleifera seeds for water treatment applications.**
3. Ali, E. N., & Muyibi, S. A. (2018). **Application of Moringa oleifera Seeds in Water Treatment.**

2.4 Agricultural Waste as Biosorbents

Agricultural waste materials such as banana peels and orange peels have attracted attention as sustainable biosorbents due to their abundance, low cost, and biodegradability. Banana peels contain cellulose, hemicellulose, lignin, and pectin, while orange peels are rich in pectin and carboxyl functional groups. These functional groups facilitate the adsorption of suspended particles, dyes, and heavy metals through ion exchange and electrostatic interactions. The utilization of agricultural waste also promotes waste valorization and supports sustainable environmental management (Crini, 2006; Mohan & Pittman, 2007).

Research Papers

1. Crini, G. (2006). **Non-conventional Low-cost Adsorbents for Dye Removal.** *Bioresource Technology.*
2. Mohan, D., & Pittman Jr., C. U. (2007). **Activated Carbons and Low-cost Adsorbents for Water Treatment.** *Journal of Hazardous Materials.*

2.5 Activated Charcoal for Water Purification

Activated charcoal is a highly porous carbonaceous material possessing a large surface area that enables efficient adsorption of dissolved organic compounds, odours, colours, and suspended impurities. Its adsorption performance depends on pore size distribution, surface chemistry, and available adsorption sites. Activated charcoal has been extensively used in water treatment due to its high adsorption capacity and compatibility with composite biomaterials (Mohan & Pittman, 2007).

Research Paper

- Mohan, D., & Pittman Jr., C. U. (2007). **Activated Carbon and Low-cost Adsorbents for Water Treatment.** *Journal of Hazardous Materials.*

2.6 Sodium Alginate Hydrogel Technology

Hydrogels are three-dimensional polymeric networks capable of retaining large quantities of water while maintaining structural stability. Sodium alginate, a naturally occurring polysaccharide obtained from brown algae, forms hydrogels when crosslinked with calcium ions through ionic interactions described by the "egg-box" model. Alginate hydrogels are widely employed for encapsulation, controlled release, and environmental applications because of their biodegradability, biocompatibility, and non-toxic nature (Rinaudo, 2008).

Research Papers

1. Rinaudo, M. (2008). **Main Properties and Current Applications of Some Polysaccharides as Biomaterials.** *Progress in Polymer Science.*
2. Lee, K. Y., & Mooney, D. J. (2012). **Alginate: Properties and Biomedical Applications.** *Progress in Polymer Science.*

2.7 Combined Coagulation and Adsorption Mechanism

Several researchers have reported that integrating coagulation and adsorption mechanisms significantly enhances purification efficiency. In such systems, Moringa proteins destabilize suspended particles through charge neutralization, whereas activated charcoal and agricultural waste adsorb contaminants through surface adsorption and biosorption. This synergistic approach improves floc formation, settling behaviour, and overall contaminant removal efficiency (Benettayeb et al., 2022).

Research Paper

- Benettayeb, A., et al. (2022). *Journal of Water Process Engineering*.

2.8 Smart Water Quality Monitoring

The integration of IoT technologies has transformed water quality monitoring by enabling real-time assessment of water parameters. ESP32 microcontrollers coupled with turbidity sensors provide an economical and efficient solution for comparative water quality analysis. Turbidity sensors operate on the principle of light scattering, where higher concentrations of suspended particles scatter more light, resulting in increased turbidity readings (Mishra et al., 2020).

Research Paper

- Mishra, A., et al. (2020). **IoT-Based Smart Water Quality Monitoring Systems: A Review.**

CHAPTER 3

SYSTEM ANALYSIS

3.1 Introduction

Water is one of the most essential natural resources, yet its quality is continuously affected by increasing levels of pollution and contamination. Suspended particles, organic pollutants, and other impurities present in water can reduce its quality and limit its usability for domestic, agricultural, and industrial purposes. Therefore, the development of sustainable and cost-effective water treatment technologies has become increasingly important. The proposed project focuses on improving water quality through the use of multifunctional biodegradable hydrogel bio-beads and a smart turbidity monitoring system. System analysis helps in understanding the existing methods, identifying their limitations, and evaluating the effectiveness of the proposed solution.

3.2 Existing System

Conventional water treatment systems generally depend on chemical coagulants and flocculants such as alum, ferric chloride, and synthetic polymers to remove suspended impurities and reduce turbidity. These treatment methods are effective but often involve high operational costs and generate chemical sludge that requires proper disposal. Continuous dependence on synthetic chemicals can also have environmental impacts and increase the complexity of the treatment process.

Most traditional systems require dedicated infrastructure and skilled personnel for operation and maintenance. In addition, they often lack real-time monitoring capabilities, making it difficult to assess water quality continuously. These limitations highlight the need for a more sustainable and economical alternative that can provide both purification and monitoring functions.

3.3 Proposed System

The proposed system utilizes biodegradable hydrogel bio-beads prepared using natural materials such as moringa powder, banana peel powder, orange peel powder, and charcoal

powder encapsulated within a sodium alginate matrix. These materials contribute to water purification through different mechanisms including coagulation, adsorption, biosorption, and chelation. The combined action of these mechanisms helps improve water clarity and reduce turbidity. The system also incorporates an ESP32 microcontroller and a turbidity sensor for real-time water quality monitoring. The sensor measures the turbidity level of the water and sends the data to the ESP32 for processing. Based on the measured turbidity, the system displays water quality information and provides visual indication through LEDs.

3.4 Feasibility Analysis

The proposed system is technically feasible because it uses simple fabrication procedures and commonly available materials. The preparation of hydrogel beads using sodium alginate and calcium chloride is straightforward and does not require sophisticated equipment. Similarly, the ESP32 microcontroller and turbidity sensor can be easily integrated into the monitoring system. From an economic perspective, the project is highly cost-effective since most of the materials used are natural and readily available. Agricultural waste products such as banana peels and orange peels can be utilized effectively, reducing material costs and promoting resource utilization.

The project is also environmentally feasible because it relies on biodegradable materials and minimizes the use of synthetic chemicals. The reuse of agricultural waste contributes to sustainable development while reducing environmental burden.

3.5 Advantages of the Proposed System

The proposed system offers several advantages compared to conventional treatment methods. The use of natural and biodegradable materials makes the process environmentally friendly and sustainable. The combination of multiple purification mechanisms improves the effectiveness of contaminant removal while reducing dependence on chemical treatment agents. Another major advantage is the integration of real-time monitoring. The ESP32-based turbidity monitoring system allows users to assess water quality instantly, eliminating the need for frequent laboratory testing. The system is also economical, easy to operate, and suitable for future enhancements and large-scale implementation.

3.6 System Architecture

The proposed system consists of two major modules: the water treatment module and the monitoring module. In the treatment module, contaminated water is treated using multifunctional hydrogel bio-beads. The active components present within the beads interact with suspended particles and contaminants, resulting in improved water quality and reduced turbidity. The monitoring module comprises a turbidity sensor connected to an ESP32 microcontroller. The sensor continuously measures the turbidity of the treated water and sends the readings to the controller. Based on the measured values, the ESP32 determines the water quality condition and activates the corresponding LED indicators. The results are also displayed through the serial monitor for further analysis.

3.7 Summary

The system analysis demonstrates that the proposed biodegradable hydrogel bio-bead system provides a sustainable, economical, and environmentally friendly approach for water quality improvement. By combining natural purification materials with smart monitoring technology, the system addresses the limitations of conventional treatment methods while offering real-time assessment of water quality. The analysis confirms the technical, economic, and environmental feasibility of the proposed solution and establishes a strong foundation for future development and implementation.

CHAPTER 4

SYSTEM SPECIFICATION

HARDWARE REQUIREMENTS

- Microcontroller : ESP32 Development Board
- Turbidity Sensor : Optical Turbidity Sensor Module (Analog)
- Power Supply : 5V DC Adapter / Power Bank
- Breadboard : Solderless Breadboard
- Jumper Wires : Male-to-Male Connecting Wires
- Voltage Divider : Two 10 k Ω Resistors (for 5V to 3.3V conversion)
- LED Indicators : Red LED, Green LED
- Resistors : 220 Ω (for LEDs)
- Hydrogel Beads : Sodium Alginate-Based Bio-Coagulant Beads
- Beaker/Container : Glass Beaker for Water Samples

4.2 SOFTWARE REQUIREMENTS

- Operating System : Windows 10/11 (64-bit)
- Programming Language : C/C++
- IDE : Arduino IDE
- Microcontroller Platform : ESP32 Board Package
- Serial Monitor : Arduino IDE Serial Monitor
- Libraries : ESP32 Arduino Core (Built-in)

CHAPTER 5

PROJECT DESCRIPTION

5.1 Problem Definition

Access to clean water remains a major challenge in many regions due to increasing levels of contamination caused by domestic waste, industrial discharge, and agricultural activities. Conventional water treatment methods often depend on chemical coagulants and expensive purification systems, making them less suitable for small-scale and rural applications. There is therefore a need for an economical, environmentally friendly, and sustainable solution capable of improving water quality while reducing dependence on synthetic chemicals.

The objective of this project is to develop multifunctional biodegradable hydrogel bio-beads using natural materials such as moringa powder, banana peel powder, orange peel powder, and charcoal powder. These bio-beads are designed to reduce turbidity and improve water quality through natural purification mechanisms. In addition, an ESP32-based turbidity monitoring system is incorporated to provide real-time assessment of water quality.

5.2 Overview of the Project

The proposed system combines biodegradable water treatment technology with smart monitoring. The treatment module consists of hydrogel bio-beads prepared using sodium alginate and calcium chloride. The active ingredients embedded within the beads interact with contaminated water and assist in the removal of suspended particles and impurities. The monitoring module consists of an ESP32 microcontroller, turbidity sensor, and LED indicators. The turbidity sensor measures the cloudiness of water and sends the data to the ESP32 for processing. Based on the measured turbidity values, the system classifies the water condition and displays the results through LED indications and serial monitor outputs.

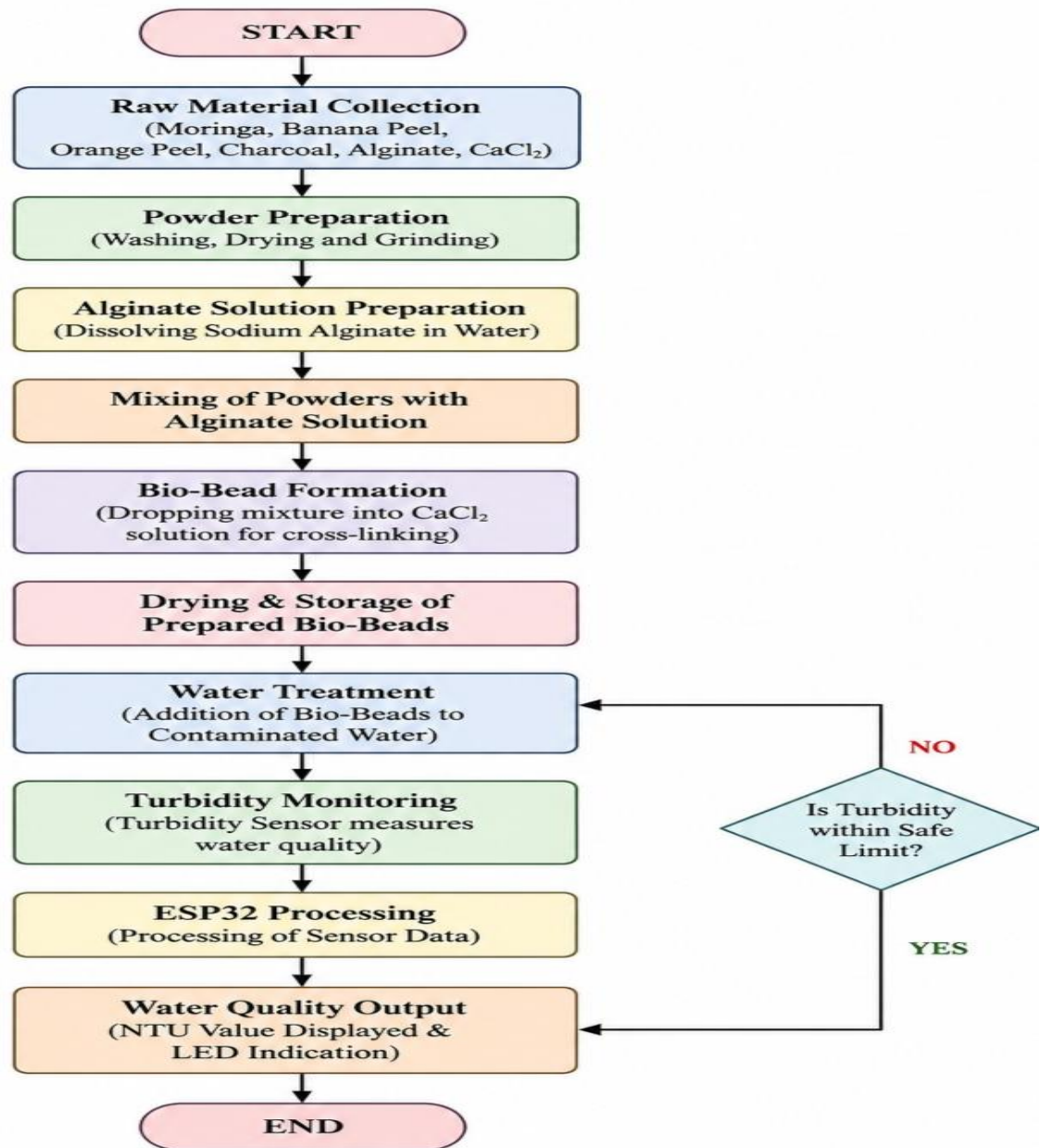


Figure 5.1: System Architecture of Proposed Project

5.3 Module Description

The project is divided into two major modules: the Water Treatment Module and the Water Quality Monitoring Module. Both modules work together to improve water quality and evaluate treatment effectiveness.

5.3.1 Modules

Module 1: Bio-Bead Preparation Module: This module involves the preparation of multifunctional hydrogel bio-beads using moringa powder, banana peel powder, orange peel powder, charcoal powder, sodium alginate, and calcium chloride. The powders are mixed with sodium alginate solution and converted into spherical beads through ionic cross-linking in calcium chloride solution.

Module 2: Water Treatment Module: In this module, the prepared bio-beads are introduced into contaminated water samples. The active components present within the beads interact with suspended particles and contaminants through adsorption, biosorption, coagulation, and chelation mechanisms. This results in improved water clarity and reduced turbidity.

Module 3: Turbidity Monitoring Module: This module consists of the ESP32 microcontroller, turbidity sensor, and LED indicators. The sensor continuously measures turbidity levels and sends the readings to the ESP32. Based on the obtained values, the controller displays the water quality condition and activates the corresponding LED indicators.

5.4 Working Principle

The working of the proposed system begins with the introduction of hydrogel bio-beads into contaminated water. Moringa powder contributes natural coagulating properties, while charcoal acts as an adsorbent for impurities. Banana peel powder and orange peel powder enhance contaminant removal through biosorption and chelation mechanisms. As treatment progresses, suspended particles begin to settle and water clarity improves. The turbidity sensor measures the change in water turbidity and transmits the data to the ESP32 microcontroller.

The processed information is displayed through the serial monitor, while LEDs provide visual indication of water quality.



Figure 5.2: Working Flow Diagram

5.5 System Design Diagram

The system design illustrates the interaction between the treatment and monitoring modules. Contaminated water is treated using multifunctional bio-beads, after which the turbidity sensor evaluates the resulting water quality. The ESP32 processes the sensor readings and provides output through LEDs and serial monitor display.

5.6 Experimental Design

The experimental setup consists of a water sample container, multifunctional bio-beads, turbidity sensor, ESP32 microcontroller, breadboard, connecting wires, and LED indicators. The contaminated water sample is treated using a specified number of bio-beads and observed over a fixed duration. Turbidity readings are recorded before and after treatment to evaluate system performance.

5.7 Input Design

The primary inputs to the system are contaminated water samples and turbidity sensor readings. Water samples containing suspended particles are used to evaluate the purification capability of the bio-beads. The turbidity sensor continuously captures turbidity values and sends them to the ESP32 for processing.

5.8 Output Design

The outputs of the proposed system include improved water clarity, reduced turbidity values, and real-time water quality indication. The turbidity sensor readings are displayed through the serial monitor, while green and red LEDs indicate low and high turbidity conditions respectively. The output enables users to assess the effectiveness of the treatment process quickly and conveniently.

CHAPTER 6

SYSTEM TESTING

6.1 Introduction

System testing was carried out to evaluate the performance of the multifunctional biodegradable hydrogel bio-beads and the turbidity monitoring system. The testing process focused on verifying bead formation, water treatment efficiency, turbidity reduction, sensor functionality, and real-time monitoring capabilities. Different experiments were conducted to ensure that each component of the system operated as intended. The results obtained during testing demonstrated the effectiveness of the proposed system in improving water quality and monitoring turbidity levels.

6.2 Test Cases

Test Case 1: Verification of Bio-Bead Formation

The prepared sodium alginate mixture containing moringa powder, banana peel powder, orange peel powder, and charcoal powder was added dropwise into calcium chloride solution. The formation of spherical and stable hydrogel beads was observed. The beads maintained their structure without disintegration, confirming the successful preparation of multifunctional bio-beads.

Test Case 2: Evaluation of Water Treatment Efficiency

The prepared bio-beads were introduced into a contaminated water sample containing suspended impurities. The water sample was observed over a period of time to study the treatment process. After treatment, a noticeable improvement in water clarity was observed, indicating that the bio-beads contributed to turbidity reduction and contaminant removal.

Test Case 3: Turbidity Sensor Testing

The turbidity sensor was tested using water samples with varying levels of turbidity. The sensor successfully detected changes in water quality and produced different output values corresponding to the turbidity levels. This confirmed the proper functioning of the sensor for water quality monitoring applications.

Test Case 4: ESP32 Monitoring System Testing

The ESP32 microcontroller was tested by interfacing it with the turbidity sensor. Sensor readings were successfully processed and displayed through the serial monitor. The system was able to continuously monitor water quality and provide real-time turbidity information without interruption.

Test Case 5: LED Indication Testing

The LED indication system was tested to verify its response to different turbidity conditions. When the turbidity level was below the predefined threshold, the green LED was activated. When the turbidity level exceeded the threshold value, the red LED was activated. The LED indicators accurately reflected the condition of the water, providing a simple visual representation of water quality.

Test Case 6: Overall System Testing

The complete system was tested by combining the bio-bead treatment process with the turbidity monitoring module. The bio-beads improved the clarity of the contaminated water, while the turbidity sensor and ESP32 continuously monitored the changes in water quality. The successful operation of all components confirmed that the integrated system met the intended project objectives. System testing was carried out to evaluate the performance of the multifunctional biodegradable hydrogel bio-beads and the turbidity monitoring system. The testing process focused on verifying bead formation, water treatment efficiency, turbidity reduction, sensor functionality, and real-time monitoring capabilities. Different experiments were conducted to ensure that each component of the system operated as intended. The results obtained during testing demonstrated the effectiveness of the proposed system in improving water quality and monitoring turbidity levels.

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 Introduction

System implementation is the stage in which the proposed system is developed, integrated, and tested to ensure proper functionality. In this project, the implementation phase involved the preparation of multifunctional biodegradable hydrogel bio-beads and the development of a smart water quality monitoring system using an ESP32 microcontroller and turbidity sensor.

The bio-beads were prepared using natural materials including moringa powder, banana peel powder, orange peel powder, charcoal powder, sodium alginate, and calcium chloride. The prepared beads were used for the treatment of contaminated water samples to evaluate their effectiveness in reducing turbidity and improving water clarity.

The monitoring module was implemented by interfacing a turbidity sensor with an ESP32 microcontroller. The sensor continuously measured turbidity levels and transmitted the readings to the controller for processing. Based on the measured turbidity values, the system displayed the water quality status and activated LED indicators to provide visual feedback. The successful integration of the treatment and monitoring modules resulted in a complete smart water quality improvement system.

7.2 Screen shots

This section presents the important stages involved in the implementation of the proposed system. The images provide a visual representation of the preparation of bio-beads, the experimental setup, the monitoring system, and the final results obtained during testing. The screenshots and photographs help in understanding the practical implementation and working of the project.

Common Name: Moringa seed powder Scientific Name: <i>Moringa oleifera</i> Major Functional Components: Cationic proteins, amino acids		Common Name: Orange Peel Scientific Name: <i>Citrus sinensis</i> Major Functional Components: Pectin, cellulose, carboxyl groups	
Common Name: Banana Peel Scientific Name: <i>Musa paradisiaca</i> Major Functional Components: Cellulose, lignin, pectin		Common Name: Activated Charcoal Major Functional Components: Porous carbon structure, high surface area, carbon micropores	

Fig. 7.1: Preparation of Raw Materials



Description: Figure 7.2 shows the preparation of the sodium alginate mixture used for bead formation. The natural powders were uniformly mixed with sodium alginate solution to obtain a homogeneous suspension suitable for bead preparation.

Fig. 7.2: Preparation of Sodium Alginate Mixture



Description: Figure 7.3 shows the formation of hydrogel bio-beads through ionic cross-linking. The alginate mixture was added dropwise into calcium chloride solution, resulting in the formation of spherical biodegradable beads.

Fig. 7.3: Formation of Hydrogel Bio-Beads



Fig. 7.4: Prepared Multifunctional Bio-Beads

Description: Figure 7.4 shows the final multifunctional hydrogel bio-beads prepared for water treatment applications. The beads contain natural materials encapsulated within the hydrogel matrix and are used for turbidity reduction experiments.



Fig. 7.5: Serial Monitor Output

Description: Figure 7.5 shows the turbidity values obtained from the monitoring system. The ESP32 continuously processed sensor readings and displayed the corresponding turbidity information through the serial monitor.



Fig. 7.7: Water Sample Before and After Treatment

Description: Figure 7.7 shows the visual comparison of the contaminated water sample before and after treatment using the multifunctional hydrogel bio-beads. The improvement in water clarity demonstrates the effectiveness of the proposed treatment system.

CHAPTER 8

CONCLUSIONS AND FUTURE ENHANCEMENTS

8.1 Conclusions

The project successfully demonstrated the development of multifunctional biodegradable hydrogel bio-beads for water quality improvement using natural and sustainable materials. The bio-beads were prepared using moringa powder, banana peel powder, orange peel powder, charcoal powder, sodium alginate, and calcium chloride. The prepared beads exhibited the ability to improve water clarity and reduce turbidity through mechanisms such as coagulation, adsorption, biosorption, and chelation.

The integration of an ESP32-based turbidity monitoring system enabled real-time assessment of water quality. The turbidity sensor successfully detected variations in turbidity levels and provided continuous monitoring through sensor outputs and LED indications. The project highlights the potential of combining biotechnology, sustainable materials, and smart monitoring systems to develop environmentally friendly water treatment solutions.

Overall, the proposed system provides a low-cost, eco-friendly, and innovative approach for water quality improvement while promoting the utilization of biodegradable materials and agricultural waste resources.

8.2 Future Enhancements

The performance of the proposed system can be further enhanced through the incorporation of additional natural adsorbents and bio-flocculants to improve contaminant removal efficiency. Future studies may focus on optimizing bead composition, porosity, and controlled release characteristics to achieve better treatment performance.

The monitoring module can be expanded by integrating additional sensors such as pH, Total Dissolved Solids (TDS), dissolved oxygen, temperature, and conductivity sensors for comprehensive water quality analysis. The implementation of IoT technology can enable remote monitoring, cloud-based data storage, and real-time access to water quality information.

Further research can also explore the application of the bio-beads for heavy metal removal, wastewater treatment, irrigation water improvement, and community-scale water purification systems. Such developments would increase the practical applicability of the project and contribute towards sustainable water resource management and environmental protection.

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APPENDIX A

A.1 Applications of the Proposed System

Household Water Pre-Treatment

The developed bio-beads can be used as a preliminary water treatment method in households to reduce turbidity and improve water clarity. The natural materials present in the beads help in the removal of suspended particles from contaminated water. This provides a simple and eco-friendly approach for improving water quality before further purification.

Rural and Remote Area Water Treatment

The bio-beads can be utilized in rural and remote regions where access to advanced water treatment facilities is limited. Their low-cost and biodegradable nature makes them suitable for community-level water purification initiatives. The system can help improve water quality using locally available and sustainable resources.

Agricultural and Irrigation Applications

The treated water can be used for irrigation purposes after turbidity reduction. Improved water quality can minimize the accumulation of suspended solids in irrigation systems and enhance water utilization efficiency. This application supports sustainable agricultural practices and water resource management.

Educational and Research Applications

The project serves as an effective educational tool for demonstrating concepts related to biotechnology, environmental engineering, water treatment, and IoT-based monitoring systems. Students and researchers can use the system to study adsorption, coagulation, and biosorption mechanisms. It also provides hands-on experience in sustainable technology development.

The bio-beads can be employed as a preliminary treatment method for wastewater containing suspended impurities. By reducing turbidity before advanced treatment processes, the overall treatment efficiency can be improved. This can contribute to reducing the operational burden on conventional wastewater treatment systems.

Environmental Monitoring Programs

When combined with the ESP32-based turbidity monitoring system, the project can be used for environmental water quality assessment. Real-time turbidity monitoring enables users to evaluate the condition of water bodies and identify changes in water quality. This can support environmental conservation and pollution control efforts.

Emergency and Disaster Relief Applications

The portable bio-bead kit can be adapted for use during emergencies, natural disasters, or temporary water shortages. It provides a lightweight and easy-to-use solution for improving the quality of available water sources. Such a system can assist affected communities where access to clean water becomes a challenge.

Smart Water Quality Monitoring Systems

The integration of bio-beads with IoT-enabled monitoring systems creates opportunities for smart water management. Continuous monitoring of turbidity levels can provide real-time feedback regarding treatment effectiveness. This approach combines sustainable purification techniques with modern sensing technologies for efficient water quality management.

A.2 Sustainable Development Goals

SDG 3 – Good Health and Well-Being

The project contributes to public health by improving water quality and reducing exposure to contaminated water sources. Cleaner water can help minimize waterborne diseases and promote healthier living conditions.

SDG 6 – Clean Water and Sanitation

The developed bio-beads support sustainable water treatment by reducing turbidity and improving water clarity. The project encourages affordable and eco-friendly approaches for enhancing water quality and sanitation.

SDG 9 – Industry, Innovation and Infrastructure

The integration of biodegradable bio-beads with an ESP32-based monitoring system demonstrates an innovative approach to water treatment. The project combines biotechnology and smart sensing technologies to address environmental challenges.

SDG 12 – Responsible Consumption and Production

Agricultural waste materials such as banana peels and orange peels are converted into value-added products instead of being discarded. This promotes resource efficiency, waste utilization, and sustainable production practices.

SDG 13 – Climate Action

The use of biodegradable and naturally derived materials reduces dependence on synthetic chemicals and supports environmentally sustainable treatment methods. This contributes indirectly to reducing environmental impact and promoting climate-friendly solutions.

A.3 Portable Bio-Bead Kit for Rural Communities

The developed Bio-Bead Kit is designed as a simple, affordable, and user-friendly water treatment solution for rural and underserved communities. The kit consists of pre-prepared biodegradable bio-beads packaged in small zip-lock pouches, making them easy to transport, store, and distribute. Since the beads are lightweight and require no electricity or specialized equipment, they can be conveniently used in areas with limited infrastructure.

The bio-beads are formulated using natural materials such as moringa, banana peel powder, orange peel powder, charcoal powder, and sodium alginate. These materials work together to reduce water turbidity through adsorption and natural treatment mechanisms. The use of biodegradable and agricultural waste-based materials makes the kit environmentally sustainable and economically feasible for large-scale deployment.

In rural regions, many communities depend on ponds, lakes, wells, and other untreated water sources that often contain suspended particles and impurities. The Bio-Bead Kit offers a low-cost treatment option that can be used before irrigation, household cleaning, or other non-potable applications. The compact packaging allows individual pouches to be distributed easily through schools, self-help groups, NGOs, and community development programs. The kit also promotes awareness regarding sustainable water treatment practices by encouraging the use of eco-friendly alternatives to conventional chemical-based methods. With further optimization and field validation, the Bio-Bead Kit has the potential to become an accessible community-level solution for improving water quality while supporting environmental sustainability and resource conservation.

